

Basic Linear Algebra (융합기초선형대수학)

KNU Math 902

Classnotes

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These notes are supplemental to a class taught from Trefethen and Bau's 'Numerical Linear Algebra'. It is expected that students have this text. It is herein referred to as 'the text'.

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The text is on Numerical Linear Algebra, and supposes a familiarity with classical linear algebra. It expects that you know what matrices and vectors are, how matrix equations are a way of writing systems of linear equations, and how we can solve them using Gaussian Elimination. You should be familiar eigenvalues and eigenvectors of a matrix, have some notion that they are important, and have seen how to find them algebraically, even if you don't remember this completely.

It doesn't matter so much if you can remember it, because in practice, we aren't going to use Gaussian Elimination. In practice, we are using big matrices, and solving them with computers, so are generally not solving them algebraically.

Gaussian Elimination is a great theoretical tool for understanding the theory of matrices, for diagonalising matrices or finding their nullspace. We will 'recall' such things, but only briefly.

We will spend much more of our time on matrix decompositions and approximations. We will look first at the QR-decomposition of a matrix, and because matrices are seldom square, we will talk more of the single value decomposition than diagonalisation.

Our focus will be on not the simplest algorithms for working with matrices, but on those that work best on computers— stable algorithms, which control their error. It is a different game than classical linear algebra, but it has its own beauty.

Lecture 1 Matrix Vector Multiplication

In this book, we work over the complex numbers $\mathbb{C}$, but it is usually okay to think of working over the reals $\mathbb{R}$. Probably all pictures, and much intuition, will be of the real case.

An $m \times n$ matrix $A = A_{m \times n}$ over the complex numbers $\mathbb{C}$ is a ‘rectangle’ of m rows and n columns of numbers from $\mathbb{C}$.

$$A = \begin{bmatrix} 1 & 2 & \frac{1}{2} \\ 4 & i & \pi \end{bmatrix}$$

is a 2×3 matrix. We write $A = [a_{i,j}]$ to designate that $a_{i,j}$ is the entry in the i^{th} row (from the top) and the j^{th} column. We drop the comma and write a_{ij} when it does not cause confusion. The values m and n are the *dimensions* of the matrix. A $1 \times n$ matrix is a (*column*) *vector* and a $m \times 1$ matrix is a (*row*) *vector*. When $A = [a_{ij}]$ we write a_{i*} for the i^{th} row vector and a_{*j} for the j^{th} column vector. The ‘ $*$ ’ runs over all values.

Vectors in $\mathbb{C}^m$ are viewed sometimes as points in $\mathbb{C}^m$, and sometimes as the ‘arrow’ from the origin $0 = (0, 0, \dots, 0)$ to the point.

Practice

Where $A = [a_{ij}]$ is the 2×3 matrix $A = \begin{bmatrix} 1 & 2 & \frac{1}{2} \\ 4 & 0 & \pi \end{bmatrix}$, what is a_{23} ? What is a_{*2} ?

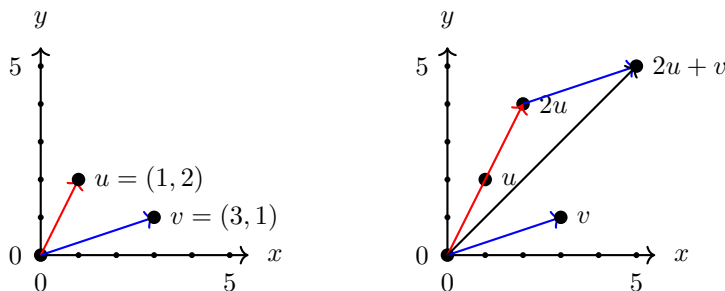
We add matrices of the same dimensions componentwise

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}$$

and multiply matrices by *scalars* $r \in \mathbb{R}$ componentwise

$$2 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 4 & 6 \end{bmatrix}.$$

For vectors (either $m \times 1$ or $1 \times n$ matrices) there is an obvious interpretation of this.



We can also multiply matrices, but only if their dimensions are right. Where $A = A_{\ell \times m} = [a_{ij}]$ and $B = B_{m \times n} = [b_{jk}]$ the product $C_{\ell \times n} = AB$ is the matrix $C = [c_{ik}]$ where

$$c_{ik} = \sum_{j=1}^m a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{im}b_{mk} = a_{i*} \cdot b_{*k}.$$

The entry $c_{ik} = a_{i*} \cdot b_{*k}$ is the dot product of the i^{th} row of A and the k^{th} column of B :

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \boxed{1} & \boxed{4} & \boxed{2} \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \boxed{3} & \cdot & \cdot \\ \cdot & \boxed{1} & \cdot & \cdot \\ \cdot & \boxed{1/2} & \cdot & \cdot \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}.$$

In the *row view of the matrix product* AB , ‘the rows of A are coefficients in linear combinations of the rows of B ’:

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 4 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \\ 2 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 3 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}.$$

In the *column view* we view the columns of B as providing coefficients in linear combinations of the columns of A .

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} & \begin{bmatrix} 7 \\ 8 \end{bmatrix} \\ \times & \times \\ \begin{bmatrix} 1 \\ 2 \end{bmatrix} & \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ + & + \\ 2 & 8 \\ \times & \times \\ \begin{bmatrix} 4 \\ 3 \end{bmatrix} & \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}$$

Practice

Compute the two products

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} a & b & c \end{bmatrix}$$

and observe that **products do not commute**– the order of the matrices is important when we are multiplying them. These are two different products of two vectors. The first is called the *inner product* or the *dot product*, and the second is called the *outer product*.

Practice

Find the 3×3 matrix A such that

$$\begin{bmatrix} 3 & 2 & 0 \\ 1 & 2 & 2 \\ 4 & 2 & 1 \end{bmatrix} = A \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Matrix times a vector

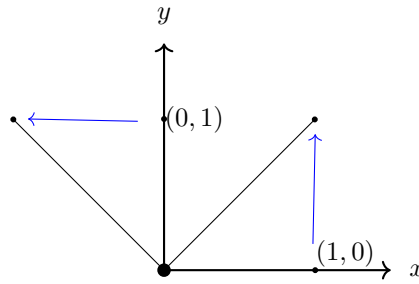
When the second matrix is a column vector $x \in \mathbb{C}^n$, multiplication

$$Ax = b$$

has two interpretations.

One interpretation is to view A as transforming the vector $x \in \mathbb{C}^n$ to a vector b in $\mathbb{C}^m$.

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



It is a linear transformation:

$$A(x + y) = Ax + Ay \quad \text{and} \quad A(\alpha x) = \alpha Ax.$$

A good way to visualise what it is doing to the other points is to look at how it transforms the unit sphere S of all points that are distance 1 from the origin. In the above example we would transform the unit circle to the circle of radius $\sqrt{2}$.

Every linear transformation from $\mathbb{C}^n$ to $\mathbb{C}^m$ can be described as multiplication by an $m \times n$ complex matrix A .

The other interpretation views $b = Ax$ as the linear combination of columns of A with coefficients given by x :

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

Range and Nullspace of A

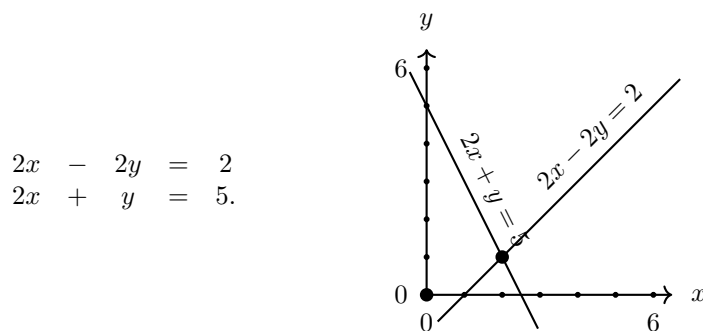
The *range*, $\text{range}(A)$, of an $m \times n$ matrix A is the set of vectors in $\mathbb{C}^m$ that can be generated as linear combinations of its columns. It is clearly a subspace of $\mathbb{C}^m$ so more commonly called the *column space* of A . The *nullspace*, $\text{null}(A)$, of A is the set of vectors x in $\mathbb{C}^n$ such that $Ax = 0$. This is also clearly a space, as $Ax = 0$ and $Ay = 0$ implies that

$$A(ax + by) = aAx + bAy = a0 + b0 = 0.$$

The range is the set of values b such that the equation $Ax = b$ has a solution and the nullspace is the set of values x such that $Ax = 0$ has a solution.

The equation $Ax = b$ and Gaussian Elimination

This is not in the text, but lets recall how we solve these equations in a traditional linear algebra course. A system of linear equations, each row of which can be viewed as a hyperplane in real space,



can be written as a matrix equation of the form $Ax = b$:

$$\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

In highschool you solved a system of equations like this. Your first step was to ‘take a copy of the first equation from the second,’ to get an equivalent system of equations.

$$\begin{array}{rcl} 2x & - & 2y = 2 \\ 2x & + & y = 5 \end{array} \rightsquigarrow \begin{array}{rcl} 2x & - & 2y = 2 \\ & & 3y = 3. \end{array}$$

The second system is the matrix equation

$$\begin{bmatrix} 2 & -2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

You should understand how to solve matrix equations in this way, using ‘row operations’ to reduce this to something like

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

from which you can easily read off the solution $(x, y) = (2, 1)$. This is called Gaussian elimination. You might have seen it done using *augmented matrices* to reduce your work:

$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right].$$

Sometimes things didn’t work so nicely. Sometimes there was no solution, and sometimes there were many. We looked at rank and determinants and inverses to characterise when $Ax = b$ has a solution.

Rank

A set $V = \{v_1, \dots, v_d\}$ of vectors in $\mathbb{C}^m$ is (*linearly*) *independent* if no one vector can be expressed as a linear combination of the others:

$$v_d = x_1 v_1 + x_2 v_2 + \dots + x_{d-1} v_{d-1};$$

equivalently, if

$$([v_1 \mid v_2 \mid \dots \mid v_d] x = 0) \rightarrow x = 0.$$

The *rank* of a matrix is the number independent columns it has. By this definition, the rank of an $m \times n$ matrix A is at most the minimum of m and n . If it is exactly $\min(m, n)$ it is *full rank*. A square $m \times m$ matrix A has full rank $\text{rank}(A) = m$ if and only if the following equivalent conditions are true.

- i. The columns of A are linearly independent.
- ii. Gaussian elimination reduces A to the identity matrix I .
- iii. A has an inverse matrix A^{-1} such that $AA^{-1} = I = A^{-1}A$.
- iv. The rows of A are linearly independent.
- v. $\text{range}(A) = \mathbb{C}^m$.
- vi. $\text{null}(A) = \{0\}$.
- vii. $\det(A) \neq 0$.
- viii. A is non-singular.
- ix. 0 is not an eigenvalue of A .

Recall that the determinant $\det(A)$ is the product of the diagonals we get after doing Gaussian elimination on A without row switches or scaling; we don't talk about it much in this class.

A square matrix is *singular* if it does not have full rank, and non-singular if it does; think of this as a definition.

A number λ is an *eigenvalue* of A if there is a non-zero solution to $Ax = \lambda x$. These we will talk about.

A matrix inverse times a vector

If we have the inverse A^{-1} of A , we can solve the equation $Ax = b$ by computing $x = A^{-1}b$. But we are usually not going to do this. Finding A^{-1} will often be computationally difficult. So the text urges you not to view $x = A^{-1}b$ as a transformation of b . They suggest that it is better viewed as a change of basis. When $A = [a_1 | a_2 | \dots | a_n]$ has independent columns a_i , $x = (x_1, \dots, x_n)$ is the vector of co-efficients x_i such that

$$b = x_1 a_1 + x_2 a_2 + \dots + x_n a_n.$$

We view $x = A^{-1}b$ as these co-efficients of b expressed in a basis of $\text{range}(A)$.

Problems from the text

Lecture 1: 1, 3, 4. (I can't understand # 2. Can anyone explain it?)

Extra for the exercises

When we were doing Gaussian elimination above we reduced

$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array} \right] \quad \text{to} \quad \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array} \right].$$

This was a row operation, we removed a copy of the first row from the second. We can view this as multiplying the first matrix, on the left, by the elementary matrix $E = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$. Under the row view of matrix multiplication, the first row of E says 'do nothing to the first row' and the second row of it says 'remove one copy of the first row from the second row':

$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array} \right] = \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array} \right].$$

Exercise 1 is asking you to start with a matrix B and do row and column operations. For the row operations, you will multiply by some matrix on the left, for the column operations, you will do so on the right. Be careful of the order.

Lecture 2 Orthogonal Vectors and Matrices

The *transpose* of a matrix $A = [a_{ij}]$ is the matrix $A^T = [a_{ji}]$ that we get by reflecting it through its diagonal. A matrix A is *symmetric* if $A = A^T$.

Recall that a complex number is of the form

$$z = a + ib$$

where $a, b \in \mathbb{R}$ and i is the positive square root of -1 . The *adjoint* of a complex number $z = a + ib$ is $\bar{z} = a - ib$. More generally the *adjoint* of a matrix $A = [a_{ij}]$ is the matrix

$$A^* = [\bar{a}_{ji}]$$

that we get from it by taking its transpose, and replacing each entry with its adjoint. If $A^* = A$ then A is *hermitian* and a real hermitian matrix is *symmetric*.

Practice

For complex numbers a and b show that $\overline{ab} = \bar{a}\bar{b} = \bar{b}\bar{a}$. Using this show for complex matrices A and B that $(AB)^* = B^*A^*$.

Most (with some definite exceptions) of the theory of adjoints and hermitian matrices goes through without adjustment to transposes and symmetric matrices when we restrict to real matrices; so many books will use A^T much more than A^* , and often use A^H for A^* . Don't be surprised if I use T occasionally, instead of $*$.

Inner product

The *inner product* of two vectors a and b in $\mathbb{C}^m$ is the product a^*b where they are viewed as column vectors. This is also called the *dot product* of the vectors, so again, don't be surprised if I write it $a \cdot b$.

Practice

Compute a^*b when $a = (2, 3, 4)$ and $b = (3, 1, -1)$. When $a = (-i, 2, 1)$ and $b = (i, i, -2)$.

A vector $v = (v_1, \dots, v_n)$ can be viewed as a point in $\mathbb{C}^n$, but also it can be viewed as the directed distance from 0 to that point. As such, its distance can be computed by the Pythagorean Theorem as

$$\|v\| = \sqrt{v^*v}.$$

A vector is *normal* if $\|v\| = 1$.

Practice

Compute $\|(3, 4)\|$.

Using the Law of Cosines, one can show that where α is the angle between vectors a and b (viewed as arrows from the origin 0), that

$$\cos \alpha = \frac{a^*b}{\|a\|\|b\|}.$$

This is 0 for non-zero a and b if and only if they are perpendicular.

Orthogonal Vectors

Non-zero vectors a and b such that $a^*b = 0$ are called *orthogonal*. Sets $X, Y \subset \mathbb{C}^m$ of non-zero vectors *orthogonal* if $x^*y = 0$ for each $x \in X$ and $y \in Y$. A set Q of non-zero vectors is *orthogonal* if $x^*y = 0$ for any two vectors $x, y \in Q$; if the vectors are also normal, then Q is *orthonormal*.

When solving $Ax = b$ we are looking to express the vector b as a linear combination of the columns of A . This is especially easy when the columns of A are a set of orthonormal vectors A .

Indeed assume that $A = [q_1 \mid \cdots \mid q_d]$ where the set $\{q_1, \dots, q_d\}$ is orthonormal. If $Ax = b$ for $x = (x_1, \dots, x_d)$, then we have that

$$b = x_1q_1 + \dots x_dq_d$$

and we can find x_i by computing

$$q_i^*b = q_i^*(x_1q_1 + \dots x_dq_d) = x_1q_i^*q_1 + \dots + x_iq_i^*q_i + \dots + x_dq_i^*q_d = x_i.$$

In particular, if there is a solution to $Ax = b$ then it is unique:

$$x = \sum_{i=1}^d (q_i^*b)q_i = \sum_{i=1}^d (q_iq_i^*)b.$$

The uniqueness gives us the following.

Fact 2.1. *A set of orthogonal vectors is independent.*

The two expressions $(q_i^*b)q_i$ and $(q_iq_i^*)b$ are the same but have different interpretations. In the first, q_i^*b is a scalar, it is the co-efficient of q_i in the expansion of b . In the second $(q_iq_i^*)$ is a matrix that projects b onto its i^{th} component.

Even if there is no solution to $Ax = b$, we can write b as

$$b = x_1q_1 + \dots x_dq_d + r$$

for some r that is orthogonal to the range of A .

Unitary Matrices

A square matrix U is *unitary* (or *orthogonal* in the real case) if its columns are orthonormal. In this case $U^*U = I$ and so $U^{-1} = U^*$. As a transformation, a unitary matrix is quite simple. It preserves the lengths of vectors and the angle between them, so is a rigid rotations of $\mathbb{C}^m$. Indeed

$$(Qu)^*(Qv) = u^*Q^*Qv = u^*Iv = u^*v$$

gives that $\|Qu\| = (Qu)^*(Qu) = u^*u = \|u\|$ and that the cosine of the angle between Qu and Qv is

$$\frac{(Qu)^*(Qv)}{\|Qu\|\|Qv\|} = (Qu)^*(Qv) = u^*v = \frac{u^*v}{\|u\|\|v\|}$$

which is the cosine of the angle between u and v .

Problems from the text

Lecture 2: 1-6. (1-4 are classical linear algebra theory.)

Lecture 3 Norms

In a classical linear algebra course we usually only consider one vector norm, the euclidean norm, because, in a real sense, all are equivalent. But numerically, other norms are often easier to work with. And when used to define matrix norms, something that is not always seen in a classical class, they have some interesting applications.

A norm is a non-negative function on vectors that has many properties similar to what you would expect of a 'length' function. Technically, a (*vector*) *norm* is a non-negative function $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}^+$ that satisfies, for all $c \in \mathbb{C}$ and $v, w \in \mathbb{C}^n$,

- $\|v\| = 0$ if and only if $v = 0$, (positive definiteness),
- $\|c \cdot v\| = |c| \cdot \|v\|$, (scaling), and
- $\|v + w\| \leq \|v\| + \|w\|$, (the triangle inequality).

For any $p \geq 1$ the function

$$\|v\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

is a norm. The most common cases we consider are when $p = 1$ or 2 , or one defined by taking $p \rightarrow \infty$:

- the ℓ^1 - or *1-norm* $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$,
- the ℓ^2 - or *Euclidean norm* $\|v\| = \|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$,
- the ℓ^∞ - or *max norm* $\|v\|_\infty = \max\{|v_1|, |v_2|, \dots, |v_n|\}$.

Problem:

Show that $\|\cdot\|_1, \|\cdot\|_2$ and $\|\cdot\|_\infty$ are norms.

Problem:

For each of the above norms $\|\cdot\|$, draw the unit sphere $S = \{v : \|v\| = 1\}$ in $\mathbb{R}^2$.

A common problem is to minimise or maximize a norm on the set of vectors in a subspace.

Problem:

Minimise the above norms on the vectors in the line $L : x/3 + y/4 = 1$.

When you do this, you will notice that the minimum point for the ℓ_1 -norm is mostly 0s in such problems. While the other norms tend to have non-zero numbers in more coordinates. Because of this the ℓ_1 -norm will often be a useful measure in approximations.

Apart from the p -norms, we also use weighted norms. For a (usually diagonal) non-singular matrix W let $\|v\|_W = \|Wv\|_2$.

Problem:

When $W = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ draw the unit sphere S for the norm $\|\cdot\|_W$.

General matrix norms

We also define norms $\|\cdot\|$ for matrices. We want that they have the same properties of positive definiteness, scaling, and triangle inequality that vector norms have, but we want that they also have sub-multiplicativity:

$$\|AB\| \leq \|A\|\|B\|.$$

Viewing an $m \times n$ matrix $A = [a_{ij}]$ as a vector of length mn :

$$(a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn})$$

we immediately get the first three properties, but we don't always get sub-multiplicativity. In the case of the 2-norm though, we do get multiplicativity by Cauchy-Schwarz, and so we do use this construction for the *Frobenius Norm*. Where $A = [a_{ij}]$ we define

$$\|A\|_F = \|(a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn})\|_2.$$

Problem

Show that the Frobenius norm satisfies sub-multiplicativity.

Matrix norms induced by vector norms

While other vector norms don't give matrix norms in this way, they do give them in another way.

For any vector norm $\|\cdot\|$ let

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{x \in S} \|Ax\|$$

be the maximum stretch of x by A in this norm. Here S is the unit sphere with respect to the given norm.

Problem:

Show that submultiplicativity is satisfied above for any choice of the vector norm $\|\cdot\|$.

To compute the norm of a matrix A induced by a vector norm, we only have to look at all vectors in the unit sphere S under that vector norm, and check which one is stretched the most by A .

In the text they draw the action of a matrix A on the unit balls under the various vector norms, and show how one can visualise the induced matrix norms from these pictures.

As for the Frobinius norm, computing the 1-norm and the ∞ -norm of a matrix are easy.

Example 3.1. To compute the 1-norm $\|A\|_1$ we for a matrix $A = [a_1 \mid \cdots \mid a_n]$ we maximise

$$\|Ax\| = x_1|a_1| + \cdots + x_n|a_n|$$

over x such that $x_1 + x_2 + \cdots + x_n = 1$. This is clearly maximised when x is the standard normal vector e_M with a 1 in the M^{th} coordinate where a_M is the largest column. So $\|A\|_1 = |Ae_M| = |a_M|$. The 1-norm of a matrix A is the maximum of the norms of its columns.

With similar analysis one can to the following.

Problem

Characterise the vectors in the unit sphere under the vector norm $\|\cdot\|_\infty$. Show that for a matrix A $\|A\|_\infty$ is the max, over all rows of A , of the sum of the entries in the row.

For other norms it is a bit harder computationally, though visually it is not so bad.

Example 3.2. Under the 2-norm, the unit sphere S is our usual unit sphere. Applying A to transform each point x in S to Ax transforms S into an paraboloid. The matrix norm $\|A\|_2$ is the length of the longest axis of this paraboloid.

Problems from the text

Lecture 3: 1-6

Background Material: Eigenvalues and Eigenvectors

An *eigenvalue* of a square matrix A is a constant λ such that

$$Ax = \lambda x$$

for some non-zero vector x . The vector x is the corresponding *eigenvector*.

Example 3.3. For the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$, we have

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

so 1 is an eigenvalue with eigenvector $(1, 0)$ and 3 is another eigenvalue with eigenvector $(0, 1)$.

For if v_1 and v_2 are both eigenvectors with the eigenvalue λ then

$$A(av_1 + bv_2) = aAv_1 + bAv_2 = a\lambda v_1 + b\lambda v_2 = \lambda(av_1 + bv_2)$$

and so the set of eigenvectors with a given eigenvalue λ are a space, called it *eigenspace*.

The matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ has eigenspaces $\langle (1, 0) \rangle$ and $\langle (0, 1) \rangle$.

Practice

Show that A has no other eigenvectors. Indeed show that an $n \times n$ matrix can have at most n eigenvalues.

In this example it was easy to find the eigenvectors by inspection. Sometimes knowing what the matrix does as a transformation can also be used to find the eigenvectors and eigenvalues.

Practice

What are the eigenvectors of the rotation $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$?

In a classical linear algebra class we would find the eigenvalues and eigenvectors of a small matrix algebraically by first finding the roots of what is called the ‘characteristic polynomial’ of the matrix. This is not practical for large matrices, so we will not do this. However there are some ideas we would get from this that are useful.

- A real matrix might have complex eigenvalues, and so complex eigenvectors.
- For a triangular matrix, the eigenvalues are the diagonal entries.

Basis Change and Similar Matrices and Diagonalisation

An $m \times m$ matrix $D = [d_{ij}]$ whose only non-zero entries are the diagonal entries d_{ii} is a *diagonal matrix*. These are quite nice to work with. As transformations they simply stretch the standard normal vector e_i by a factor of d_{ii} . When we transform the unit sphere $S = \{v \in \mathbb{C}^m \mid \|v\| = 1\}$ by D we get a hyperellipsoid DS whose i -axis has length exactly d_{ii} .

A more general $m \times m$ matrix A might not transform S as nicely. It generally has some rotating and some stretching. A change of basis can often transform all vectors so that the action of A is nicer. To understand the action of A we transform the vectors, do the nice action, and then transform them back. Algebraically this looks like writing A as $A = BXB^{-1}$ for an invertible matrix B . The action Ax of A on a vector x decomposes into $BXB^{-1}x$ where B^{-1} transforms the vector, X does the nice action and then B transforms us back to the original setting.

Note

I find change of basis one of the most difficult to explain 'easy' idea. It is personal, if you don't understand my explanation, try the one in the book. Try other books.

A diagonalisation of A is the nicest example of a change of basis transformation. If $Ax = \lambda x$ and B is invertible, then for the vector Bx we have

$$BAB^{-1}(Bx) = BAx = B\lambda x = \lambda Bx.$$

So A and BAB^{-1} have the same eigenvalue λ . (Its eigenvector under A is x , and its eigenvector under BAB^{-1} is Bx .) Two $m \times m$ matrices are *similar* if they have the same eigenvalues.

If A has a full set of eigenvectors, that is, m independent eigenvectors $\{v_1, \dots, v_m\}$, then we have

$$A[v_1 | v_2 | \dots | v_m] = [\lambda_1 v_1 | \dots | \lambda_m v_m] = [v_1 | v_2 | \dots | v_m] \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix}.$$

Letting X be the matrix whose i^{th} column is the i^{th} eigenvector v_i and Λ be the diagonal matrix whose i^{th} entry is λ_i , we get the *diagonalisation*

$$A = X\Lambda X^{-1}$$

of A .

Problem

Show that where A has diagonalisation $A = X\Lambda X^{-1}$, and S is the unit sphere, AS is the hyperellipse that has radius λ_i in the direction of v_i , for all i .

The diagonalisation has some nice applications. We can easily compute powers A^n of A :

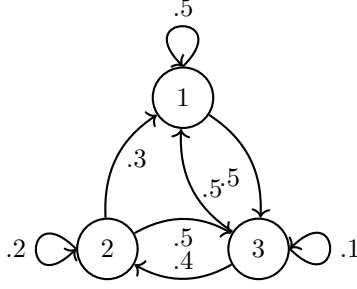
$$A^p = (X\Lambda X^{-1})^p = X\Lambda^p X^{-1} = X \begin{bmatrix} \lambda_1^p & & & \\ & \lambda_2^p & & \\ & & \ddots & \\ & & & \lambda_n^p \end{bmatrix} X^{-1}.$$

We use this in the following example.

Example 3.4. The matrix

$$T = \begin{bmatrix} .5 & .3 & .5 \\ 0 & .2 & .4 \\ .5 & .5 & .1 \end{bmatrix}$$

is the transition matrix for the Markov process described by the following state diagram:



There are three states: 1, 2 and 3. A *state* vector $u(t) = (p_1(t), p_2(t), p_3(t))$ lists the probability $p_i(t)$ that we are in state i at time t . The state digram, and the transition matrix T , show the probability t_{ij} that, if in state i at time t , we will be in state j at time $t + 1$.

So

$$u(t+1) = Tu(t),$$

and for some initial state $u(0)$ we have that

$$u(t) = T^t u(0),$$

To 'solve' the Markov Process, we want to find the limiting state vector $u(t)$ as t goes to infinity.

Well we can find the eigens of T :

$$\begin{aligned} \lambda_1 &= -2/3 & v_1 &= (1, 2, -3) \\ \lambda_2 &= 1/5 & v_2 &= (1, -1, 0) \\ \lambda_3 &= 1 & v_3 &= (13, 5, 10) \end{aligned}$$

We notice that if $p(0) = \frac{1}{28}(13, 5, 10)$ our initial state has eigenvalue one, and so we stay in that state for ever. But what happens if we start in some other state?

To see, we first diagonalise T :

$$T = S\Lambda S^{-1} = \frac{1}{84} \begin{bmatrix} 1 & 1 & 13 \\ 2 & -2 & 5 \\ -3 & 0 & 10 \end{bmatrix} \begin{bmatrix} -2/3 & & \\ & 1/5 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 10 & 10 & -18 \\ 35 & -49 & -21 \\ 3 & 3 & 3 \end{bmatrix}.$$

Then we can compute $u(t) = T^t u(0) = X\Lambda^t X^{-1}u(0)$:

$$\begin{aligned} u(t) &= \begin{bmatrix} 1 & 1 & 13 \\ 2 & -2 & 5 \\ -3 & 0 & 10 \end{bmatrix} \begin{bmatrix} -2/3 & & \\ & 1/5 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 5/42 & 5/42 & -3/14 \\ 5/12 & -7/12 & -1/4 \\ 1/28 & 1/28 & 1/28 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 13 \\ 2 & -2 & 5 \\ -3 & 0 & 10 \end{bmatrix} \begin{bmatrix} -2/3 & & \\ & 1/5 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 5p_1/42 + 5p_2/42 - 3p_3/14 =: c_1 \\ 5p_1/12 - 7p_2/12 - 1p_3/4 =: c_2 \\ 1p_1/28 + 1p_2/28 + 1p_3/28 =: c_3 \end{bmatrix} \\ &= c_1(-2/3)^t \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + c_2(1/5)^t \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + c_3(1)^t \begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix} \end{aligned}$$

Now as t gets big, the first two disappear, and so

$$u_t \rightarrow c_3 \begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix} = \frac{1}{28} \begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix},$$

which is our steady state. We don't even have to compute c_3 , as we know that this must be a state vector, so we can scale it to make it one. The initial position $u(0)$ didn't even matter. The other two terms go to 0, and so c_3 is non-zero and we will definitely converge to the state $\frac{1}{28}(13, 5, 10)$ that we got by normalising the largest eigenvector.

Lecture 4 The Singular Value Decomposition

In a classical linear algebra class we concentrate on the eigenvalues λ_i and eigenvectors v_i of a square matrix A —the objects such that

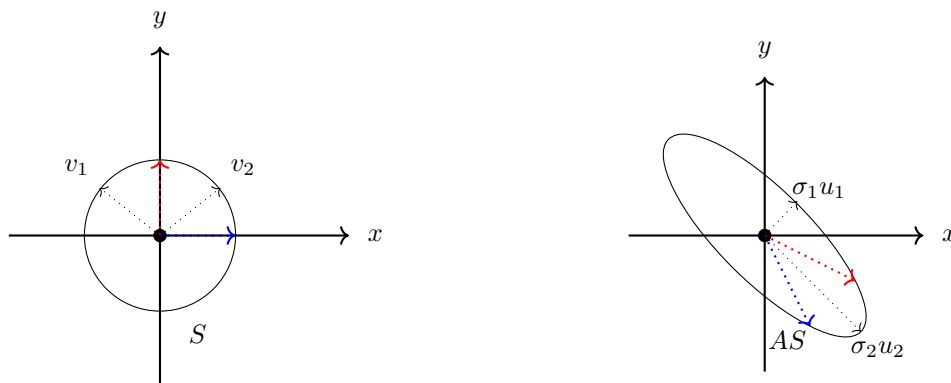
$$Av_i = \lambda_i v_i.$$

We use these to (hopefully) diagonalise the matrix. When the matrix is symmetric these eigenvalues would all be non-negative reals. When there are all positive, then the matrix has a full set of orthogonal eigenvectors, and we have a diagonalisation of the matrix by an orthogonal matrix. We have a lot of nice applications in this case, and the action of the matrix as a transformation is transparent. It maps the unit sphere to a hyperellipse, the eigenvector of the largest eigenvalue becoming the main axis.

This is a quick recall of a lot of theory, but we are not going to cover this. We aren't concentrating on square matrices, our matrices often don't have eigenvalues; and even when they are square, we might not have a full set of eigenvalues.

Singular values are a bit more forgiving, they always exist, even for non-square matrices, and can always be taken to be orthogonal. We present them by starting from the notion of the hyperellipse.

Let A be an $m \times n$ matrix. As a transformation it takes vectors in $\mathbb{C}^n$ to $\mathbb{C}^m$. For this discussion, it is probably easier to think of it as a real matrix, taking vectors in $\mathbb{R}^n$ to $\mathbb{R}^m$. As it is linear, we can interpolate its action on the unit sphere S in $\mathbb{R}^n$ by computing its action on the standard normal vectors. This is perhaps more intuitive when $m > n$, and A has full rank with $m > n$, but it is not too hard to see that the action of A takes S in $\mathbb{R}^n$ to some hyperellipse AS in $\mathbb{R}^n \subset \mathbb{R}^m$.



To describe this hyperellipse, we need its axes $\sigma_i u_i$, and to describe the transformation we also need to know the vectors v_i of S such that $Av_i = \sigma_i u_i$.

The values σ_i , which are usually ordered in non-increasing order

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n,$$

are the *singular values* of A . The vectors $u_1, \dots, u_n$ are the *left singular vectors*, and the vectors

$v_1, \dots, v_n$ are the *right singular vectors*. As $Av_i = \sigma_i u_i$ for each i we have that

$$A [v_1 \mid \dots \mid v_n] = [u_1 \mid \dots \mid u_n] \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix},$$

and so writing these matrices as $V, \hat{U}$ and $\hat{\Sigma}$, where V and $\hat{U}$ will both turn out to be orthogonal matrices, so, in particular $V^{-1} = V^*$, we get that

$$A = \hat{U} \hat{\Sigma} V^*.$$

This is called the *reduced singular value decomposition (reduced SVD)* of A . The hats on $\hat{U}$ and $\hat{\Sigma}$ are because though this is the one we use more, there is a 'full SVD' which we write without hats. The order of the matrices in the decomposition explain the use of 'left' and 'right' for the names of the special vectors.

Indeed, specifying the dimensions of the matrices, we have

$$A_{m \times n} = \hat{U}_{m \times n} \hat{\Sigma}_{n \times n} V_{n \times n}^*.$$

Padding the bottom of $\hat{\Sigma}$ with rows of 0 so that it is $\Sigma_{m \times n}$ we can extend $\hat{U}$ to U by adding $m - n$ more orthogonal columns, and so make U square, and so unitary (orthogonal in the real case), too. Doing so we get the full SVD decomposition

$$A = U \Sigma V^* = U_{m \times m} \Sigma_{m \times n} V_{n \times n}^*$$

of A — because the extra rows in Σ are all 0, the added columns of U do not change the product.

If $m < n$ then we pad $\hat{\Sigma}$ with extra columns of 0, and V^* with extra rows.

Formal Definition

As for square matrices, we call an $m \times n$ matrix $A = [a_{ij}]$ diagonal if the only non-zero entries are the a_{ii} .

A full SVD of an $m \times n$ matrix A is a factorisation $A = U \Sigma V^*$ such that U and V^* are unitary, and Σ is diagonal with non-negative real numbers $\sigma_1 \geq \dots \geq \sigma_{\min(m,n)}$ on the diagonal.

The following is proved in the text.

Theorem 4.1. *Every complex matrix A has a full SVD. Further, the matrix Σ is unique, and if it is square and the σ_i are distinct, then U and V^* are uniquely determined upto scalar factors of absolute value 1.*

Problems from the text

Lecture 4: 1-5 (For question 3, use any programming language.)

Lecture 5 More on the SVD

As for a diagonalisation $A = X\Lambda X^{-1}$ of a square matrix, the SVD $A = U\Sigma V^*$ of a general matrix A can be viewed as a change of basis. But after changing to the nice basis, doing the diagonal action, we don't change back to the original basis, we finish off by changing to some orthogonal basis of the co-domain of A as a transformation.

We lose the fact that we end up in the standard normal basis in the codomain as would naturally occur by viewing A as a transformation, but we do end up in an orthonormal basis defined by U . And for small relaxation, we get that *any* matrix has a diagonalisation.

Matrix Properties via the SVD

Recalling that the rank of a matrix is the number independent columns, or the number of independent rows that it has, we get the following.

Theorem 5.1. *The rank of a matrix is the number of non-zero singular values it has.*

Proof. We know that the rank of a product of matrices is the minimum of the ranks of the matrices. In the SVD $A = U\Sigma V^*$ the matrices U and V^* both have full rank, m and n respectively, and the rank of sigma Σ is at most the min of these, and being diagonal, is equal to the number of non-zero entries. $\square$

Sometimes it is difficult to see what A does as a transformation, even when we compute it on the standard normal vectors $e_1, \dots, e_m$. Any invertible $m \times m$ matrix B can be used as a basis change.

Much like a diagonalisation of a square matrix, the SVD allows us to change our bases, and view the action of a matrix A as diagonal action.

An $m \times n$ matrix $A = U\Sigma V^*$ is linear transformation

$$A : \mathbb{C}^n \rightarrow \mathbb{C}^m : v \mapsto Av.$$

A similarity transformation of a square matrix A , conjugating it as BAB^{-1} for some basis change matrix B